

Lab Experiments

Course Code/Course Name: DSA0604/Data Handling and Visualization

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Set 6: Product Sales Analysis

Code:

```
library(ggplot2)
library(tidyr)

product_data <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January_Sales = c(2000,1500,1200),
  February_Sales = c(2200,1800,1400),
  March_Sales = c(2400,1600,1100)
)

# Convert data to long format
sales_long <- pivot_longer(
  product_data,
  cols = c(January_Sales, February_Sales, March_Sales),
  names_to = "Month",
  values_to = "Sales"
)

sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January_Sales","February_Sales","March_Sales"),
  labels = c("January","February","March")
)

# 1. Grouped Bar Chart
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Quarterly Product Sales",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```

2. Stacked Area Chart

```
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```

3. Table

```
print(product_data)
```

```
library(ggplot2)
```

```
library(plotly)
```

```
product_data <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January_Sales = c(2000,1500,1200),
  February_Sales = c(2200,1800,1400),
  March_Sales = c(2400,1600,1100)
)
```

```
library(tidyr)
```

```
sales_long <- pivot_longer(
  product_data,
  cols = c(January_Sales, February_Sales, March_Sales),
  names_to = "Month",
  values_to = "Sales"
)
```

```
sales_long$Month <- factor(
  sales_long$Month,
  levels = c("January_Sales","February_Sales","March_Sales"),
  labels = c("January","February","March")
)
```

Interactive Grouped Bar Chart

```
p1 <- ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Quarterly Product Sales")
```

```
ggplotly(p1)
```

```
# Interactive Stacked Area Chart
```

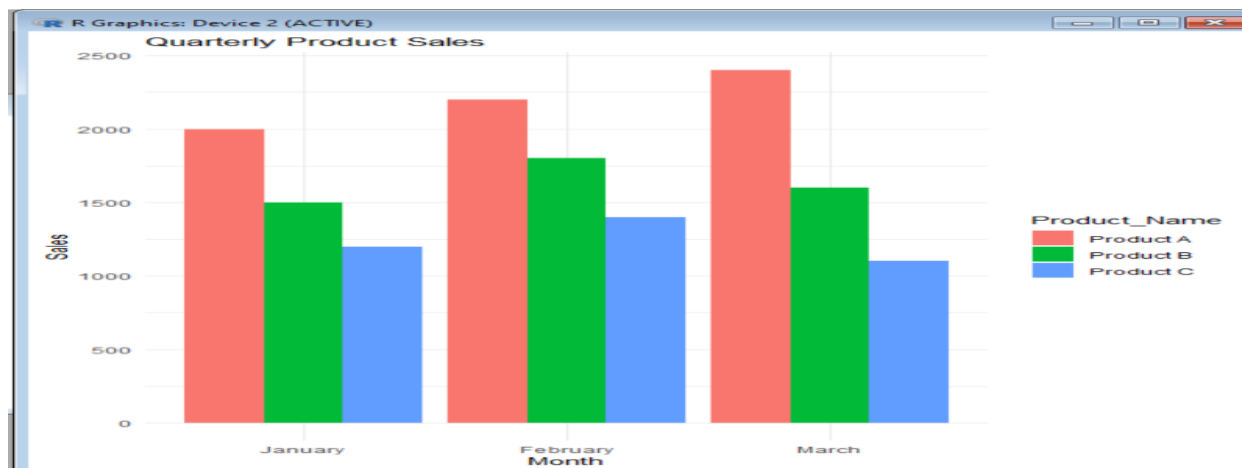
```
p2 <- ggplot(sales_long,  
  aes(x = Month,  
    y = Sales,  
    fill = Product_Name,  
    group = Product_Name)) +  
  geom_area() +  
  labs(title = "Overall Sales Trend")
```

```
ggplotly(p2)
```

```
# Sales Table
```

```
print(product_data)
```

Output:

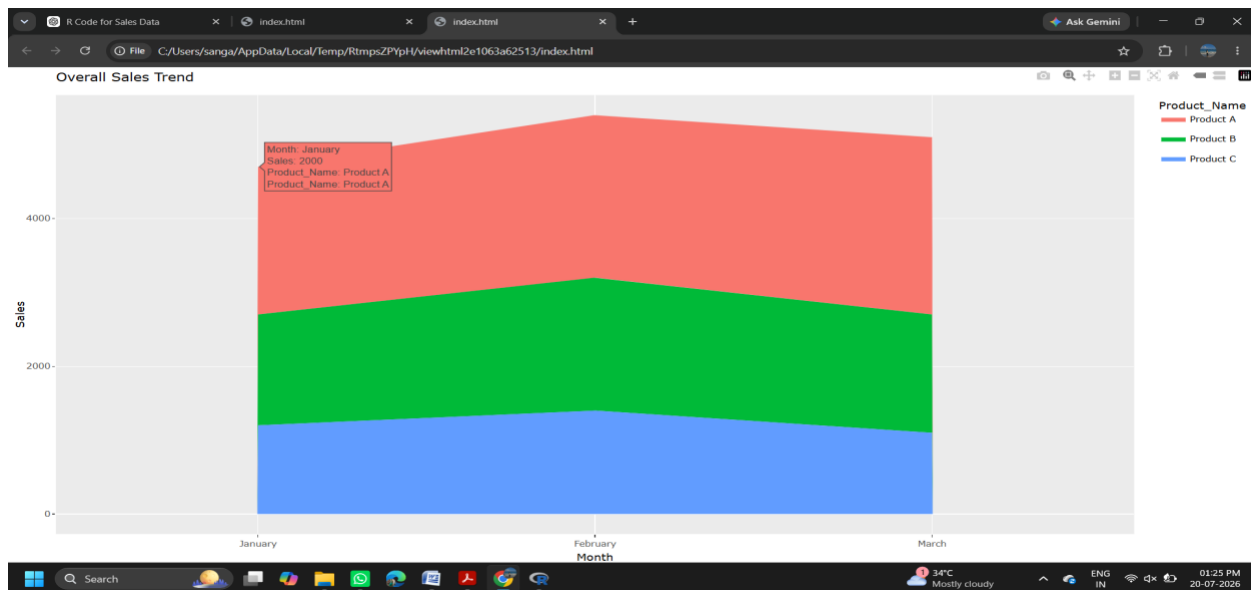
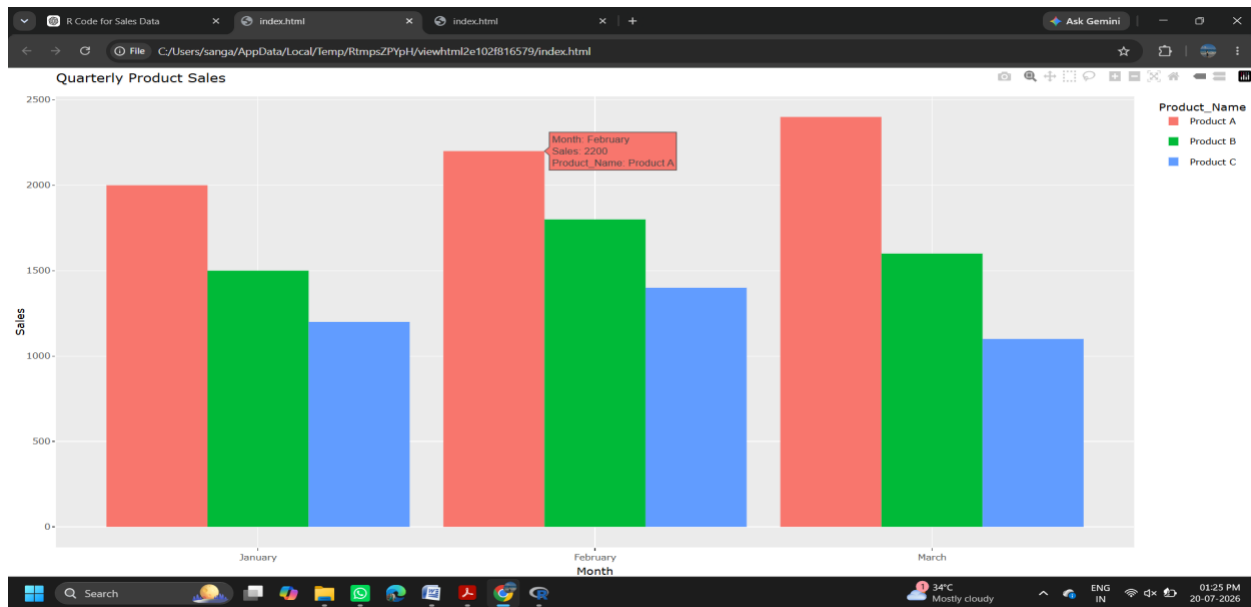


```
R Console
+ y = "Sales"
+ ) +
+ theme_minimal()
>
> # 2. Stacked Area Chart
> ggplot(sales_long,
+   aes(x = Month,
+     y = Sales,
+     fill = Product_Name,
+     group = Product_Name)) +
+   geom_area() +
+   labs(
+     title = "Overall Sales Trend",
+     x = "Month",
+     y = "Sales"
+   ) +
+   theme_minimal()
>
> # 3. Table
> print(product_data)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A           2000           2200           2400
2          2   Product B           1500           1800           1600
3          3   Product C           1200           1400           1100
>

Untitled - R Editor
y = Sales,
fill = Product_Name)) +
geom_bar(stat = "identity", position = "dodge") +
labs(
  title = "Quarterly Product Sales",
  x = "Month",
  y = "Sales"
) +
theme_minimal()

# 2. Stacked Area Chart
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()

# 3. Table
print(product_data)
```



Set 7: Customer Demographics Analysis

R code:

```
library(ggplot2)
```

```
library(plotly)
```

```
customer_data <- data.frame(
```

```
  Customer_ID = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

1. Bar Chart

```
ggplot(customer_data,  
  aes(x = factor(Age),  
    fill = factor(Age))) +  
geom_bar() +  
labs(  
  title = "Distribution of Customer Ages",  
  x = "Age",  
  y = "Number of Customers"  
) +  
theme_minimal()
```

2. Pie Chart

```
gender_count <- table(customer_data$Gender)  
pie(  
  gender_count,  
  labels = names(gender_count),  
  col = rainbow(length(gender_count)),  
  main = "Customer Distribution by Gender"  
)
```

3. Table

```
print(customer_data)
```

4. Interactive Dashboard

Interactive Bar Chart

```
p1 <- ggplot(customer_data,  
  aes(x = factor(Age),  
    fill = factor(Age))) +  
geom_bar() +  
labs(title = "Distribution of Customer Ages")
```

```
ggplotly(p1)
```

```
# Interactive Pie Chart
```

```
pie_data <- data.frame(  
  Gender = names(gender_count),  
  Count = as.numeric(gender_count)  
)
```

```
p2 <- plot_ly(  
  pie_data,  
  labels = ~Gender,  
  values = ~Count,  
  type = "pie"  
)
```

```
p2
```

```
# Interactive Table
```

```
View(customer_data)
```

Output:



```

+ geom_bar() +
+ labs(
+   title = "Distribution of Customer Ages",
+   x = "Age",
+   y = "Number of Customers"
+ ) +
+ theme_minimal()
>
> # 2. Pie Chart
> gender_count <- table(customer_data$Gender)
>
> pie(
+   gender_count,
+   labels = names(gender_count),
+   col = rainbow(length(gender_count)),
+   main = "Customer Distribution by Gender"
+ )
>
> # 3. Table
> print(customer_data)
  Customer_ID Age Gender Income
1           1  28 Female  50000
2           2  35 Male    60000
3           3  42 Female  75000
>

```

```

main = "Customer Distribution by Gender"
)

# 3. Table
print(customer_data)

# 4. Interactive Dashboard

# Interactive Bar Chart
p1 <- ggplot(customer_data,
  aes(x = factor(Age),
      fill = factor(Age))) +
  geom_bar() +
  labs(title = "Distribution of Customer Ages")

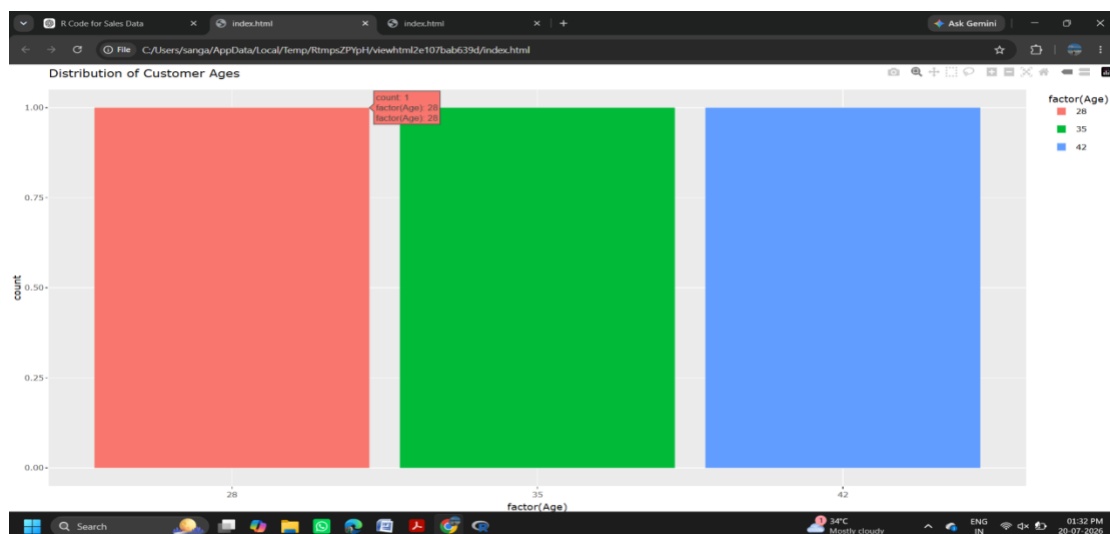
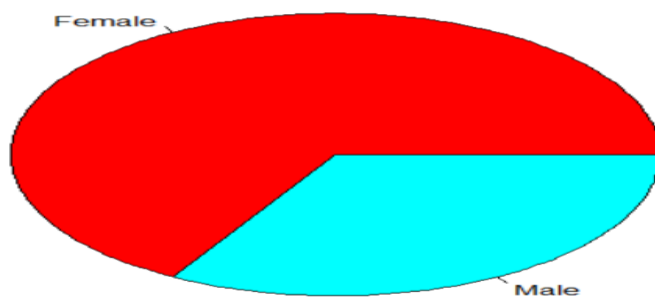
ggplotly(p1)

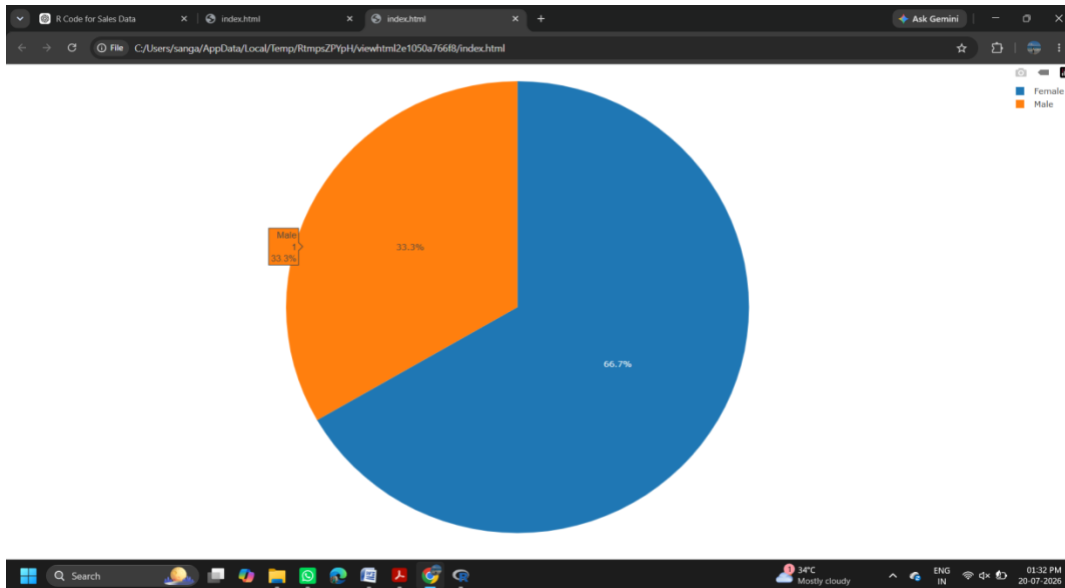
# Interactive Pie Chart
pie_data <- data.frame(
  Gender = names(gender_count),
  Count = as.numeric(gender_count)
)

p2 <- plot_ly(
  pie_data,
  labels = ~Gender,

```

Customer Distribution by Gender





Set 8: Employee Performance Analysis

R code:

```
library(ggplot2)
library(plotly)
employee_data <- data.frame(
  Employee_ID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  Years_of_Service = c(5,3,7),
  Performance_Score = c(85,92,78)
)

# 1. Line Chart
ggplot(employee_data,
  aes(x = Employee_ID,
    y = Performance_Score,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
```



```
x = "Employee ID",  
y = "Performance Score"  
) +  
theme_minimal()
```

2. Bar Chart

```
ggplot(employee_data,  
  aes(x = Department,  
    fill = Department)) +  
geom_bar() +  
labs(  
  title = "Employees Across Departments",  
  x = "Department",  
  y = "Number of Employees"  
) +  
theme_minimal()
```

3. Table

```
print(employee_data)
```

4. Interactive Dashboard

Interactive Line Chart

```
p1 <- ggplot(employee_data,  
  aes(x = Employee_ID,  
    y = Performance_Score,  
    group = 1)) +  
geom_line(color = "blue") +  
geom_point(color = "red", size = 3) +  
labs(title = "Employee Performance Trend")  
ggplotly(p1)
```

Interactive Bar Chart

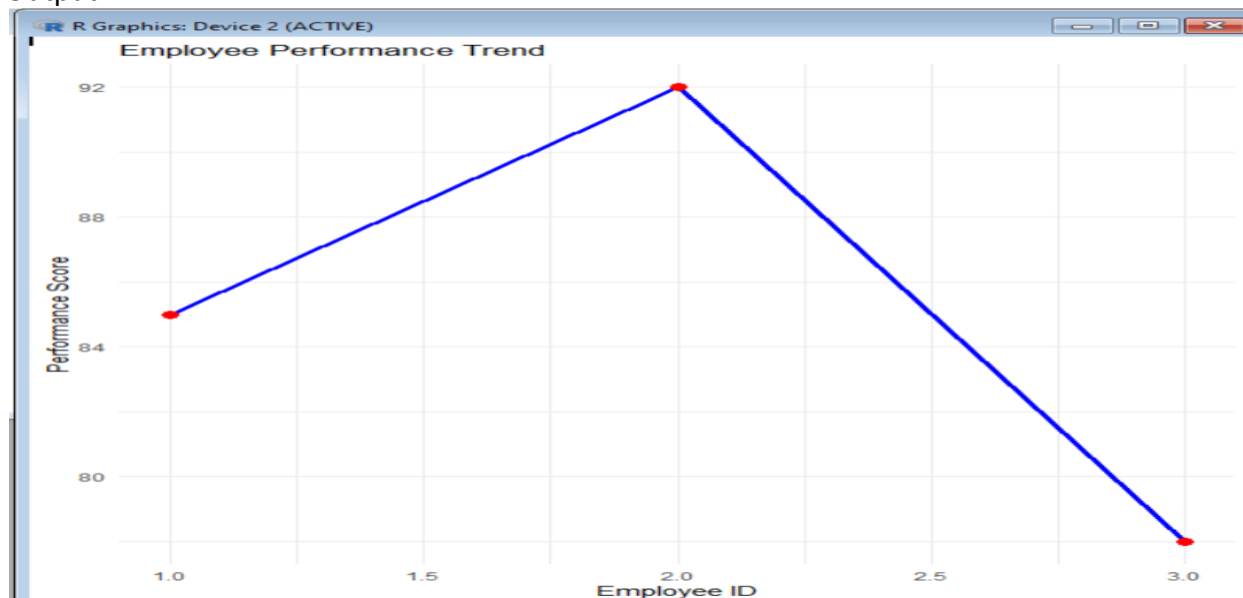
```
p2 <- ggplot(employee_data,
```

```

aes(x = Department,
    fill = Department)) +
geom_bar() +
labs(title = "Employees Across Departments")
ggplotly(p2)
# Interactive Table
View(employee_data)
...

```

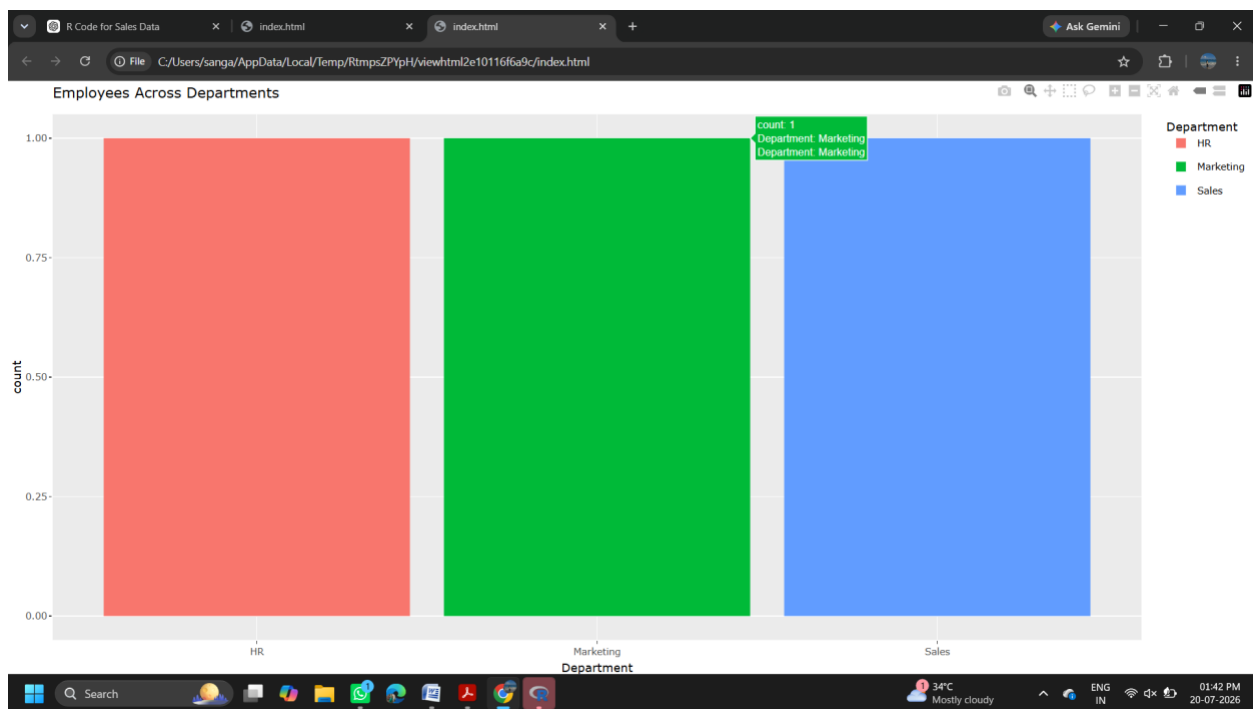
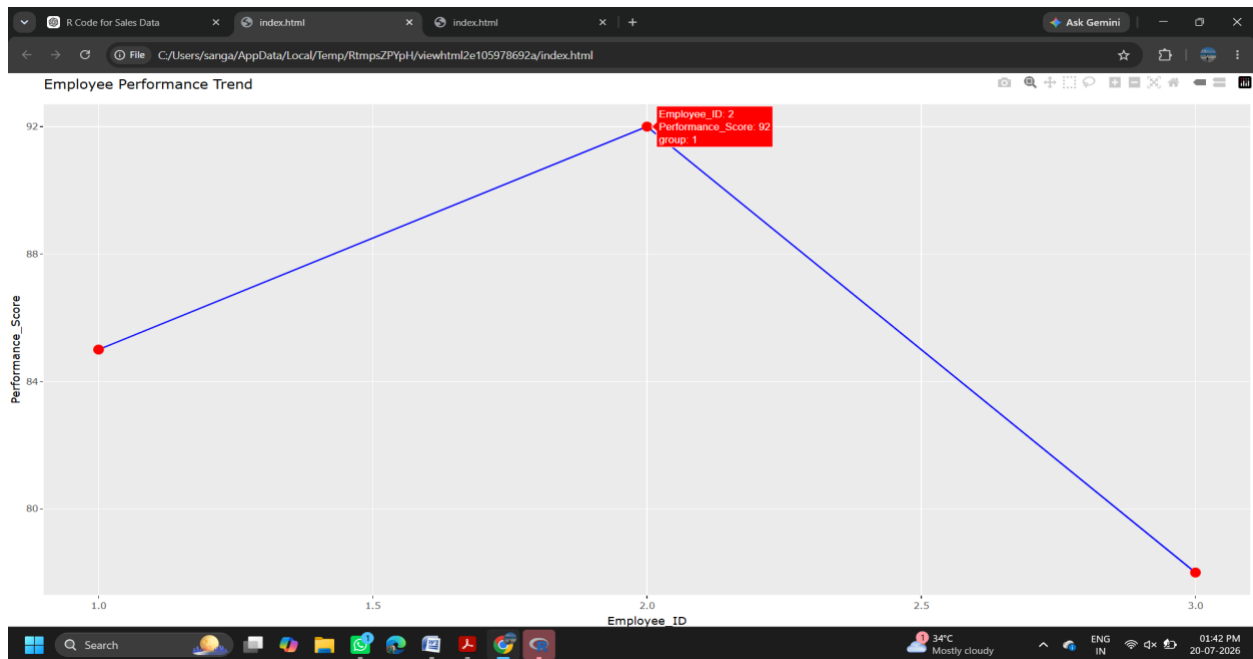
Output:



```

> print(employee_data)
  Employee_ID Department Years_of_Service Performance_Score
1           1     Sales              5             85
2           2        HR              3             92
3           3 Marketing              7             78
>
> |

```



R Data: employee_data

	Employee_ID	Department	Years_of_Service	Performance_Score
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

Set 9: Online Learning Activity

R code:

```
library(ggplot2)
```

```
library(plotly)
```

```
library(zoo)
```

```
# Dataset
```

```
data <- data.frame(  
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),  
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),  
  Age = c(20, 22, 19, 21, 23, 20),  
  Course = c("R", "R", "SQL", "R", "R", "SQL"),  
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),  
  Videos_Watched = c(12, 15, 8, 18, 9, 14),  
  Quiz_Score = c(78, 85, 65, 92, 70, 88),  
  Login_Date = as.Date(c("2025-01-05", "2025-01-05", "2025-02-08",  
    "2025-02-08", "2025-03-12", "2025-03-12"))  
)
```

```
# 1. Histogram and Boxplot
```

```
ggplot(data, aes(x = Quiz_Score)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(title = "Quiz Score Distribution",  
    x = "Quiz Score",  
    y = "Frequency")
```

```
ggplot(data, aes(x = Course, y = Quiz_Score, fill = Course)) +  
  geom_boxplot() +  
  labs(title = "Quiz Score by Course",  
    x = "Course",  
    y = "Quiz Score")
```

```
# 2. Scatter Plot
```

```
ggplot(data,  
  aes(x = Study_Time,  
    y = Quiz_Score,  
    size = Videos_Watched)) +  
  geom_point(alpha = 0.7, color = "blue") +  
  labs(title = "Study Time vs Quiz Score",  
    x = "Study Time (hrs)",  
    y = "Quiz Score")
```

```
# Question 3
```

```

library(ggplot2)

data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Login_Date = as.Date(c("2025-01-05", "2025-01-05", "2025-02-08",
    "2025-02-08", "2025-03-12", "2025-03-12"))
)

# Extract Month
data$Month <- format(data$Login_Date, "%Y-%m")

# Calculate Monthly Average Quiz Score
avg_score <- aggregate(Quiz_Score ~ Month, data = data, FUN = mean)

print(avg_score)

# Line Chart
ggplot(avg_score, aes(x = Month, y = Quiz_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Average Monthly Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +
  theme_minimal()

#4library(ggplot2)
library(plotly)

data <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Course = c("R", "R", "SQL", "R", "R", "SQL"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Quiz_Score = c(78, 85, 65, 92, 70, 88)
)

# Interactive Bar Chart
p1 <- ggplot(data,
  aes(x = Course,
    y = Quiz_Score,

```

```

    fill = Course)) +
  stat_summary(fun = mean, geom = "bar") +
  labs(
    title = "Average Quiz Score by Course",
    x = "Course",
    y = "Average Quiz Score"
  )

```

```
ggplotly(p1)
```

```
# Interactive Scatter Plot
```

```

p2 <- ggplot(data,
  aes(x = Study_Time,
    y = Quiz_Score,
    color = Gender)) +
  geom_point(size = 4) +
  labs(
    title = "Study Time vs Quiz Score",
    x = "Study Time (hrs)",
    y = "Quiz Score"
  )

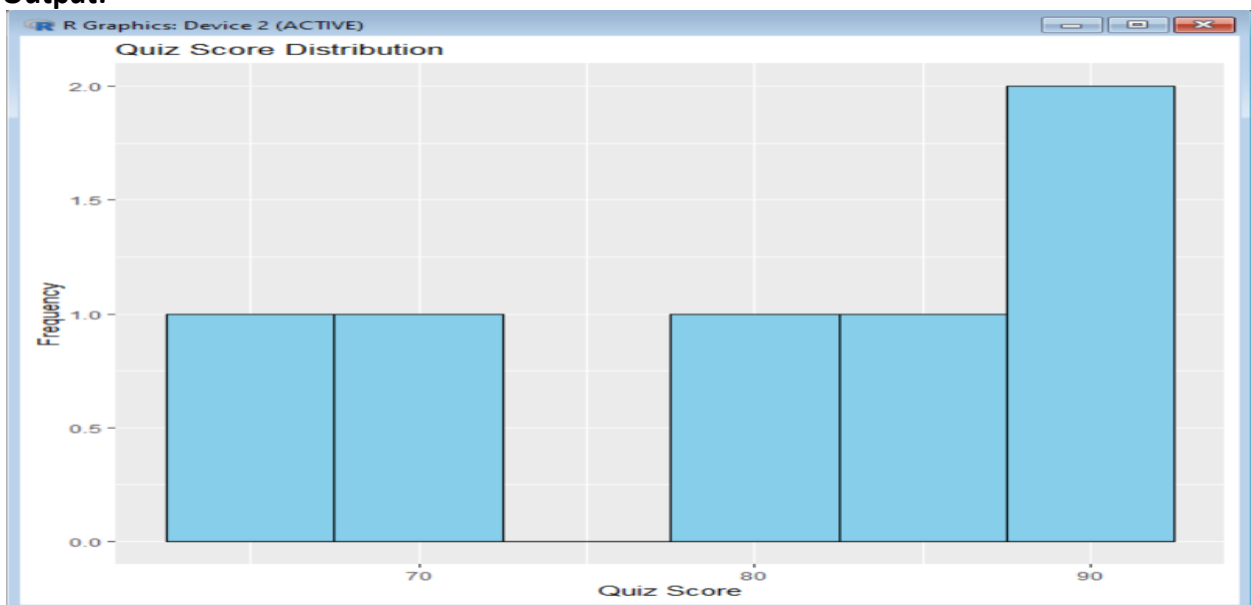
```

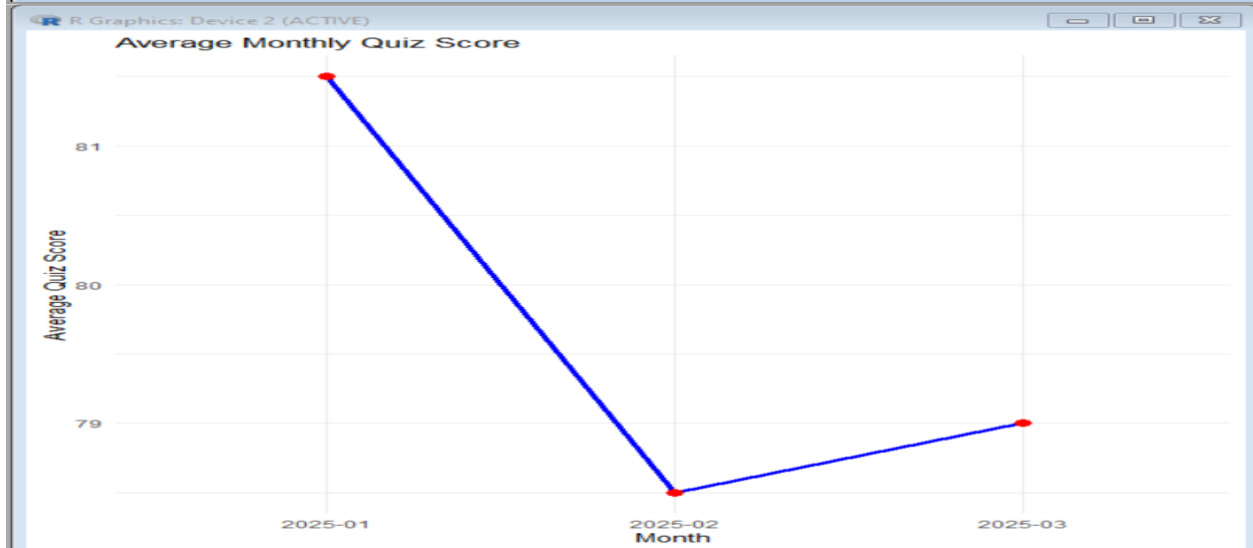
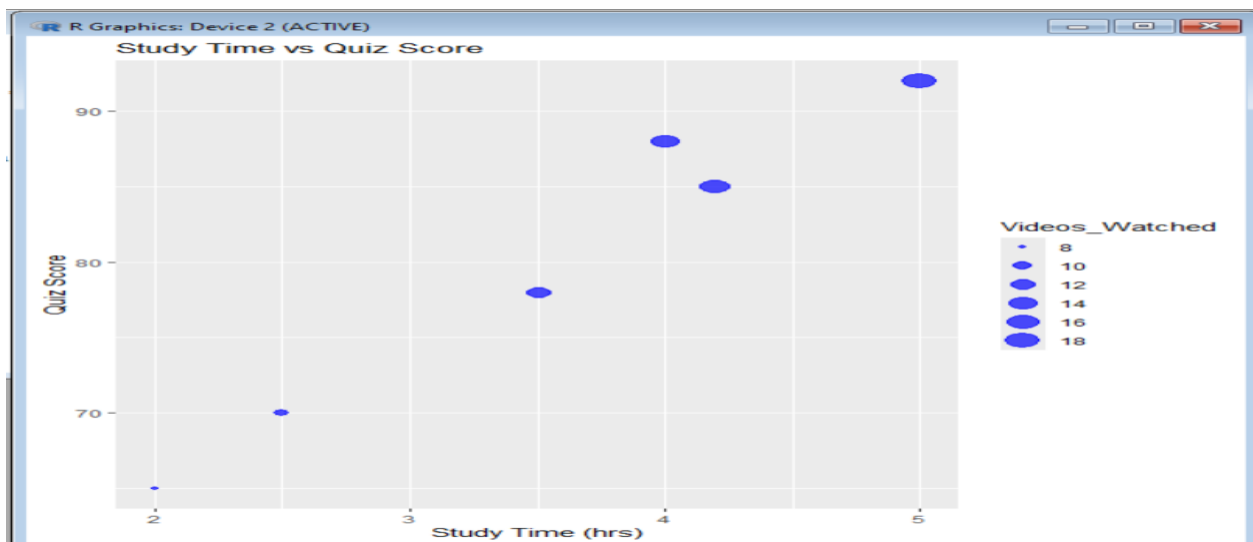
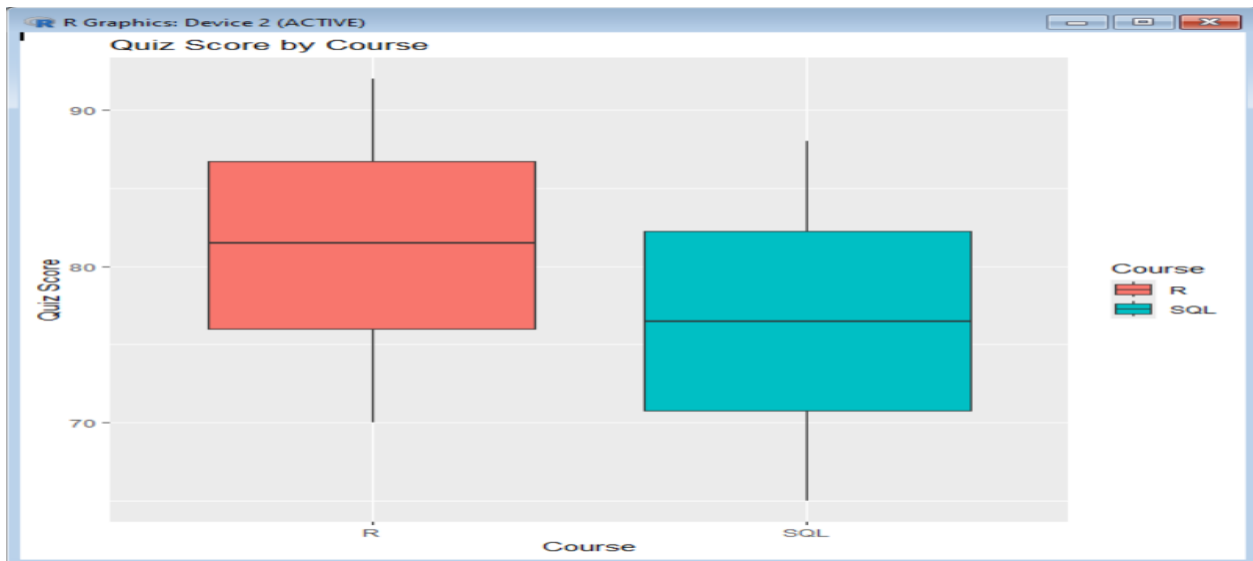
```
ggplotly(p2)
```

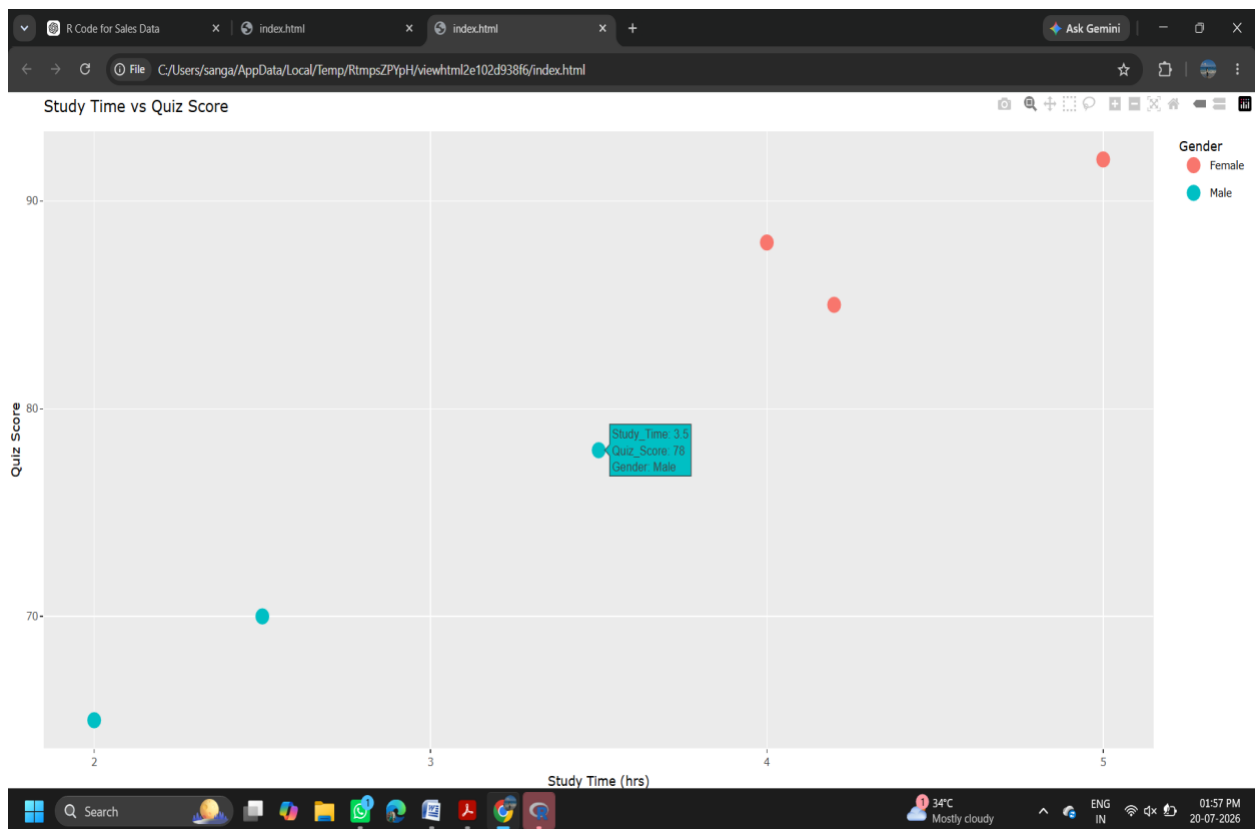
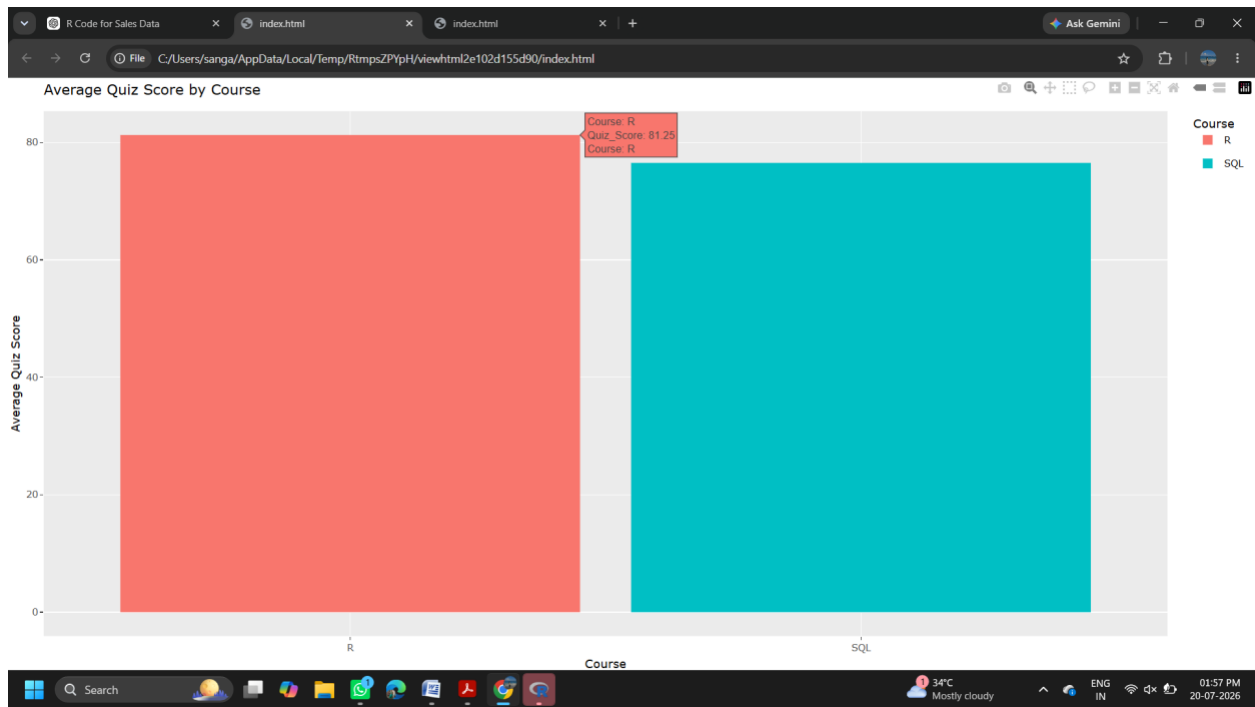
```
# Display Dataset
```

```
View(data)
```

Output:







Set 10: Survey Responses Analysis

R code:

```
library(ggplot2)
```

```
library(plotly)
```

```
library(tidyr)
```

```
survey_data <- data.frame(  
  Survey_ID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)
```

1. Grouped Bar Chart

```
q1_count <- as.data.frame(table(survey_data$Question1))  
colnames(q1_count) <- c("Answer","Count")
```

```
ggplot(q1_count, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity", position = "dodge") +  
  labs(  
    title = "Distribution of Answers for Question 1",  
    x = "Answer",  
    y = "Count"  
  ) +  
  theme_minimal()
```

2. Stacked Bar Chart

```
survey_long <- pivot_longer(  
  survey_data,  
  cols = c(Question1, Question2, Question3),  
  names_to = "Question",  
  values_to = "Answer"  
)
```

```
ggplot(survey_long,  
  aes(x = Question,  
    fill = Answer)) +  
  geom_bar() +  
  labs(  
    title = "Overall Survey Responses",  
    x = "Question",  
    y = "Count"  
  ) +
```

```
theme_minimal()
```

```
# 3. Table
```

```
print(survey_data)
```

```
# 4. Interactive Dashboard
```

```
p1 <- ggplot(q1_count,  
  aes(x = Answer,  
    y = Count,  
    fill = Answer)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Question 1 Responses")
```

```
ggplotly(p1)
```

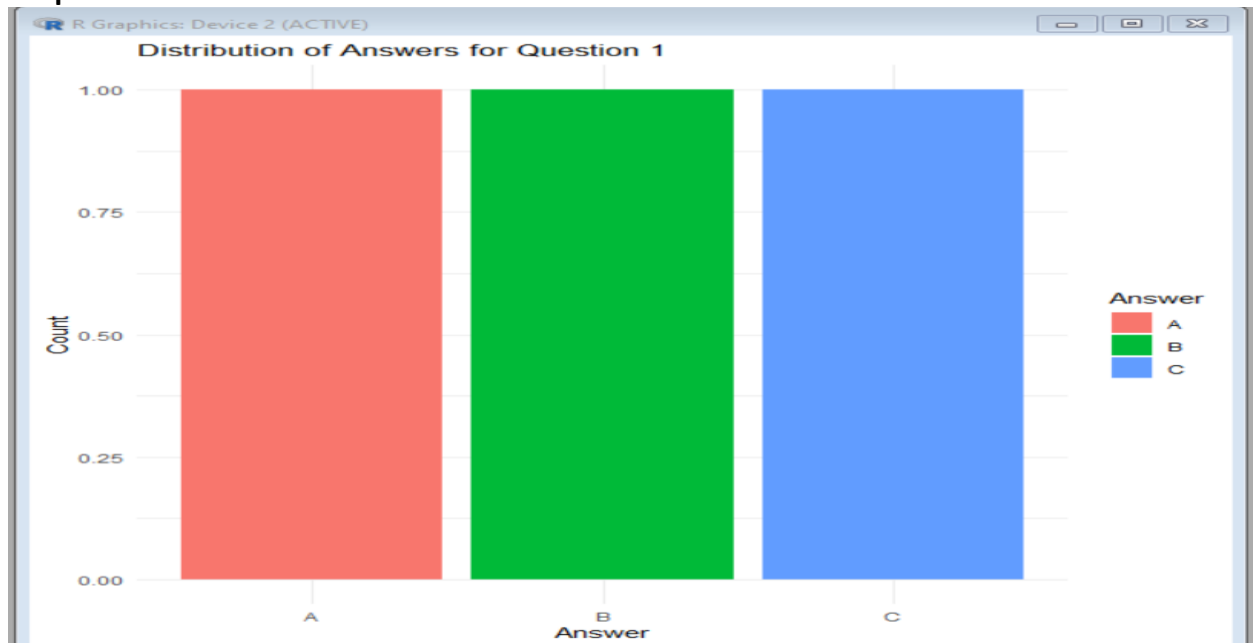
```
p2 <- ggplot(survey_long,  
  aes(x = Question,  
    fill = Answer)) +  
  geom_bar() +  
  labs(title = "Overall Survey Responses")
```

```
ggplotly(p2)
```

```
View(survey_data)
```

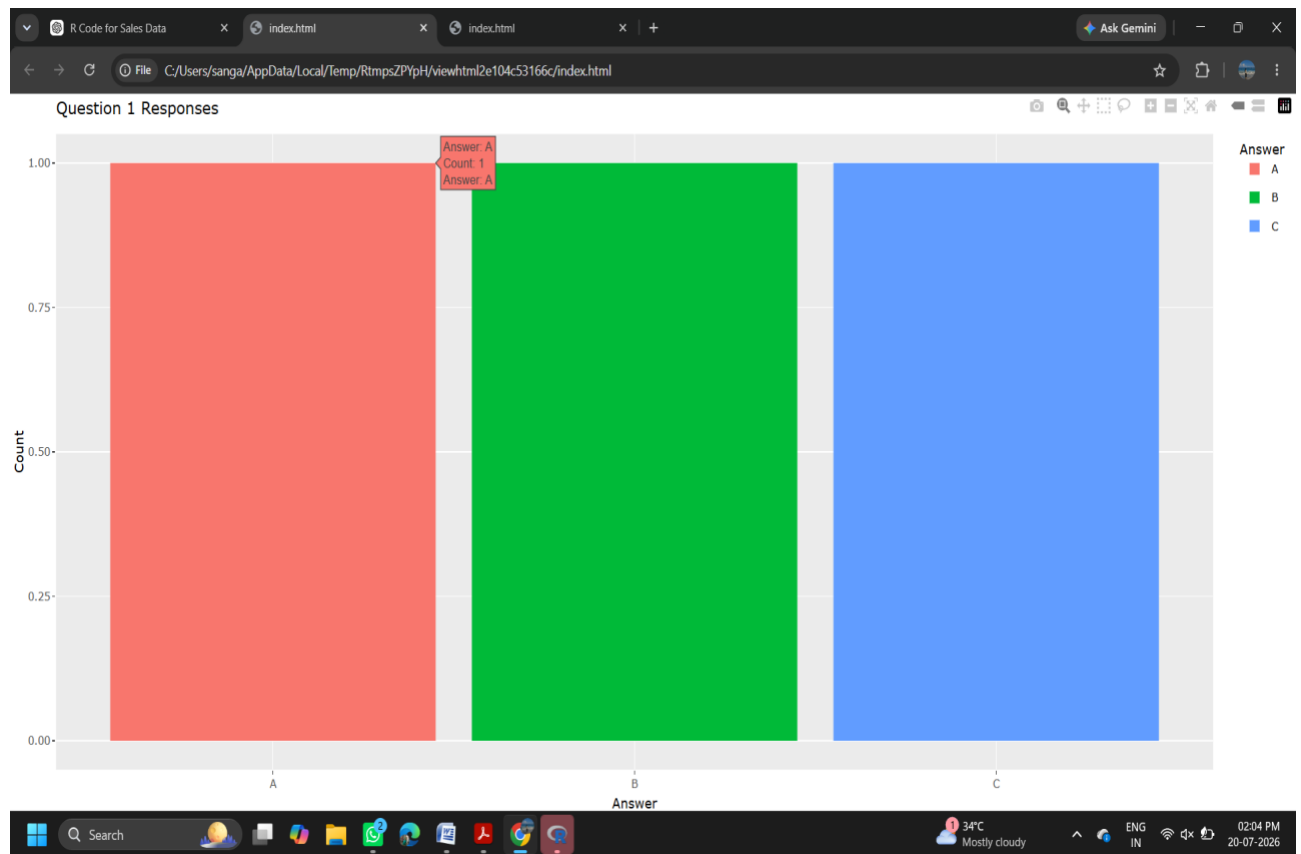
```
...
```

Output:





```
> print(survey_data)
  Survey_ID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
> |
```



Data: survey_data

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B